VT-447

Laminate/Prepreg

UL Approval: E214381 Version: C1 /127 /128 /130

General Information

- Halogen Free & High Tg (175°C) FR4.1
- CAF Resistance
- Low Z-CTE
- Excellent Thermal Reliability

IPC® Validation Services LISTED

Application

Mobile Phone, Smart Phone, Computer, Communication Equipment, Instrumentation, Electronic Game Machine, VCR, etc.

Availability

- Core Thickness: 0.002" (0.05mm) to 0.200" (5mm), available in sheet or panel form
- Copper Foil: 1/4oz to 12oz
- Prepregs are available in roll or panel form
- E-Glass styles: 7628, 1506, 1500, 2113, 2313, 3313, 2116, 1080, 1086, 1078, 106 &1067, etc

Note: For cores ≤ 0.005", it is recommended to use the reverse treated copper due to the low profile. The peel strength for RT foil is ≈1-2lbs/in (0.35Kg/m) less than standard foil.

Properties		Prepreg		Laminate
Storage Condition	Temperature	Below 23°C (73°F)	Below 5°C (41°F)	Room
	Relative humidity	Below 55% RH	/	/
Shelf Life		3 months	6 months	24 months (airproof)

The prepreg exceeding shelf life should be retested.

CORPORATE HEADQUARTERS
Ventec Electronics Co. Ltd.
308 TaiShan Rd,
New District Suzhou,
Jiangsu, P.R. China 215129
T: +86 512-68091810
Email: sales@ventec.com.cn

EMEA REGIONAL HEADQUARTERS
Ventec Europe
Unit 1 Trojan Business Centre,
Tachbrook Park Estate,
Leamington-Spa, CV34 6RH, UK
T: +44 1926-889822
Email: sales@ventec-europe.com

AMERICAS REGIONAL HEADQUARTERS
Ventec USA
720 Lee St., Elk Grove Village,
Illinois, IL 60007,
United States of America
T: +1 630-4221627
Email: saleseast@ventec-usa.com

www.venteclaminates.com

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IPC-4101E /127 /128 /130

Properties		Test Method	Units	Specification	Typical Value
Thermal Properties					
Tg	DSC	IPC-TM-650 2.4.25	°C	–	–
	TMA	IPC-TM-650 2.4.24	°C	170 minimum	175
Td		ASTM D3850	°C	340 minimum	385
T260		IPC-TM-650 2.4.24.1	Minute	30 minimum	>60
T288		IPC-TM-650 2.4.24.1	Minute	15 minimum	>30
Thermal Stress @ 288°C		IPC-TM-650 2.4.13.1	Second	Pass 10s	>600
Z-axis CTE	Before Tg	IPC-TM-650 2.4.24	ppm/°C	60 maximum	38
	After Tg	IPC-TM-650 2.4.24	ppm/°C	300 maximum	185
	Total Expansion (50~260°C)	IPC-TM-650 2.4.24	%	3.0 maximum	2.1
X-Y CTE		IPC-TM-650 2.4.24	ppm/°C	–	11~13
MOT		UL 94	°C	–	130
Electrical Properties					
DK @ 1GHz RC 50%		IPC-TM-650 2.5.5.9	–	5.4 maximum	4.35
DF @ 1GHz RC 50%		IPC-TM-650 2.5.5.9	–	0.035 maximum	0.013
Volume Resistivity	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ-cm	1.0E+4 minimum	5.0E+8
	E-24/125	IPC-TM-650 2.5.17.1	MΩ-cm	1.0E+3 minimum	5.0E+6
Surface Resistivity	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ	1.0E+4 minimum	5.0E+7
	E-24/125	IPC-TM-650 2.5.17.1	MΩ	1.0E+3 minimum	5.0E+6
Electrical Strength		IPC-TM-650 2.5.6.2	Volt/mil (KV/mm)	762 (30) minimum	1200~1400 (54)
Dielectric Breakdown		IPC-TM-650 2.5.6	KV	40 minimum	60
Comparative Tracking Index (CTI)		ASTM D3638	Rating (Volt)	–	Grade 3 (175-250)
Arc Resistance		ASTM D495	Second	–	195
Mechanical Properties					
Peel Strength (1oz)	As received	IPC-TM-650 2.4.8	lb/in (N/mm)	–	7~9 (1.2~1.6)
	After thermal stress	IPC-TM-650 2.4.8	lb/in (N/mm)	6 (1.05) minimum	7~9 (1.2~1.6)
Flexural Strength	Warp	IPC-TM-650 2.4.4	KPsi (MPa)	60 (415) minimum	72 (500)
	Fill	IPC-TM-650 2.4.4	KPsi (MPa)	50 (345) minimum	58 (400)
Physical Properties					
Moisture Absorption		IPC-TM-650 2.6.2.1	%	0.80 maximum	0.18
Flammability		UL-94	Rating	V-0 minimum	V-0

Note: All test data provided are typical values and are not intended to be specification values.

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Ventec Europe
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Tachbrook Park Estate,
Leamington-Spa, CV34 6RH, UK
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United States of America
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